

The linguexx package

Numbered examples, interlinear glosses and grammaticality judgments
for linguistics

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Abstract

`linguexx` typesets the standard apparatus of linguistic writing: numbered examples with lettered sub-examples, grammaticality judgments that hang in the margin, interlinear morpheme-by-morpheme glosses with any number of tiers, cross-references that follow the numbering automatically, and examples inside footnotes. Its input syntax is deliberately terse — `\ex.` starts an example, a blank line ends it — so that data does not disappear under markup.

That syntax is not new, and no claim is made that it is: it is the syntax of Wolfgang Sternefeld’s `linguex`, which this package owes almost everything to, and which it is designed to replace. `linguexx` is a reimplementaion from the ground up, undertaken for two reasons. The first is to repair the structural weaknesses of the original: a globally drifting sub-example depth counter (the source of the notorious **Check example numbering (screwed up?)**), a judgment tokenizer built on catcode manipulation, and a termination rule that failed at the end of a `beamer` frame. All three are gone — not patched, but designed out. The second is to bring the glossing up to the level of John Frampton’s `expex`, the one facility in which `linguex` was decisively behind: glosses may now have any number of tiers, each with its own font, rather than the two or three that `cgloss4e` allowed.

The package’s dependencies are `graphicx` and `tikz` (used only by `\alt`), plus the `expl3` layer, part of the `LATEX` kernel since 2020, and it works identically under `pdfLATEX`, `XYLATEX` and `LuaLATEX`. This manual is self-contained and assumes no acquaintance with `linguex` or any other example package: section 1 gets you writing examples in two pages; sections 3–7 treat each facility in turn; section 9 is a command reference. Section 2 sets out the relation to `linguex`, `expex` and `gb4e` for those who know them.

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1 A two-page tour

Load the package:

```
| \usepackage{linguexx}
```

An example is introduced by `\ex.` — note the period, which is part of the command name — and ended by a *blank line*:

```
| \ex. Colourless green ideas sleep furiously.
```

renders as:

(1) Colourless green ideas sleep furiously.

Numbering is automatic and runs through the document. Sub-examples are introduced by `\a.`, and continued by `\b.`, `\c.` and so on:

```

\ex.
\ a. Ich habe geschlafen.
\ b. Ich bin geschlafen.

```

renders as:

- (2) a. Ich habe geschlafen.
 b. Ich bin geschlafen.

A grammaticality judgment typed at the start of an example is recognized automatically and set in the margin, so that the example texts stay aligned:

```

\ex.
\ a. Ich habe geschlafen.
\ b. *Ich bin geschlafen.

```

renders as:

- (3) a. Ich habe geschlafen.
 b. *Ich bin geschlafen.

Interlinear glosses are written with `\gll` (two tiers), with the free translation on a `\glt` line. The tiers are aligned word by word, and category labels are set in small capitals with `\lpzg`:

```

\ex. \gll Ich schlief. \
      I      sleep.\lpzg{pst} \
\glt 'I slept.'

```

renders as:

- (4) Ich schlief.
 I sleep.PST
 'I slept.'

To refer to an example, label it and use `\ref`, exactly as with any other L^AT_EX counter; the parentheses come for free. `\Next` and `\Last` refer to the next and the previous example without a label:

```

\ex.\label{ex:sleep} Ich habe geschlafen.

As \ref{ex:sleep} shows, ... The contrast in \Next\ is sharper:

\ex. *Ich bin geschlafen.

```

renders as:

- (5) Ich habe geschlafen.

As (5) shows, ... The contrast in (6) is sharper:

- (6) *Ich bin geschlafen.

That is the whole of everyday use. The rest of this manual fills in the details: more nesting levels, more gloss tiers, arbitrary judgment marks, reference ranges, examples in footnotes, source attributions, and the layout parameters.

2 Origins, and relation to other packages

Readers who have never used another example package can skip this section with no loss; it is here because the debts are real and ought to be stated plainly.

linguex (Wolfgang Sternefeld). The input language of `linguexx` *is* the input language of `linguex`, deliberately and almost exhaustively: `\ex.`, `\a.-\f.`, `\z.`, termination by blank line, custom labels in brackets, automatic roman numbering in footnotes, the `\Next`/`\Last` family, `\exg.` and its relatives, and the layout parameters down to their names. Existing `linguex` documents compile essentially unchanged; the default geometry is a little tighter than `linguex`'s, and the option `legacy` reproduces the original spacing exactly (section 8). What has been replaced is the machinery underneath. `linguex` tracks sub-example depth in a global counter that drifts out of step whenever an example is malformed, and the damage then propagates to the numbering of every later example; it recognizes judgments with a hand-written tokenizer that manipulates category codes; and it scans an example body up to the next `\par`, so that an example may not end a `beamer` frame. In `linguexx` the depth is held in the grouping structure itself, so drift is not merely unlikely but impossible; judgments are recognized by token inspection with no catcode trickery, and any mark whatever may be used (section 4); and the body is collected by a scanner that stops at a blank line, at `\z.`, or at a structural boundary such as `\end{frame}`. Two facilities of `linguex` were dropped as more trouble than they are worth: examples embedded inside sub-examples, and the index variants `\exi./\ai..`

cgloss4e (Hans-Peter Kolb and Craig Thiersch). The glossing commands `\gll`, `\glll` and `\glt` come from here, by way of `linguex`. The engine behind them is new, and the vertical space that `cgloss4e` inserted around a gloss is gone by construction.

expex (John Frampton). `expex` is the most capable glossing package in the field, and on that one axis `linguex` was simply outclassed: two or three tiers against an arbitrary number. `\gl ... \endgl` (section 5) closes that gap — any number of tiers, each with its own font command. It does not close every gap: `expex` retains finer control over the vertical layout of individual tiers, and its keyval interface exposes more parameters than the eight lengths of section 8. If the glossing is the hard part of your document, `expex` may still be the better instrument.

gb4e. Not an ancestor of the implementation, but its environment syntax is accepted: `exe`, `xlist` and the bracketed judgment form of `\ex` work as they do there (section 3.5). It is worth knowing that `gb4e` makes `_` and `^` active in text mode, which is a frequent source of clashes with other packages; `linguexx` changes no category codes.

Parenthesis-free references. The `\pref`/`\pNext` family follows a construction of Alan Munn's.

3 Examples

3.1 Starting and ending an example

`\ex.` starts a numbered example. Three things end it:

1. a **blank line** (or an explicit `\par`) — the normal case;

2. `\z.` (section 3.3), which ends it at the point where it stands;
3. a **structural boundary**: the end of the enclosing environment or group. An example may therefore run directly into `\end{<frame>}` or the closing brace of a group, with no blank line at all.

Environments *inside* an example do not end it: a `tabular`, a `tikzpicture` or a math display is passed through untouched.

3.2 Sub-examples

`\a.` opens a level of sub-examples; `\b.` through `\f.` add further items at the level currently open. The letters printed come from a counter, not from the command name, so `\b.` prints “b.” only because it happens to be the second item — the commands are interchangeable and purely mnemonic. A second `\a.` opens a deeper level, numbered in roman:

```
\ex.
\a. First.
\b. Second.
\a. First of the deeper level.
\b. Second of the deeper level.
\c. Back at the letter level? No -- see below.
```

renders as:

- (7)
- a. First.
 - b. Second.
 - i. First of the deeper level.
 - ii. Second of the deeper level.
 - iii. Back at the letter level? No – see below.

The last `\c.` comes out as item iii: the continuation commands stay at the level currently open. Two sub-levels are the maximum; a third `\a.` raises an error. To *leave* a level, use `\z..`

3.3 Returning to a higher level: `\z.`

`\z.` pops exactly one level, counting the surrounding prose as the outermost level. From the roman level it returns to the letters, so that a following `\b.` continues there; from the letter level — or in an example with no sub-examples open — it ends the example, and what follows is ordinary prose. Consecutive `\z.`’s therefore pop successively out of the example:

```
\ex.
\a. First
\b. Second
\a. Second a
\b. Second b
\z.
\b. third
\z.
Ordinary text again.
```

renders as:

- (8) a. First
 b. Second
 i. Second a
 ii. Second b
 c. third

Ordinary text again.

The first `\z.` leaves the roman level, so `\b.` yields item c; the second leaves the example. Text on the same line as an example-ending `\z.` — like “Ordinary text again” above — is a *continuation*: it is set flush left under the example, without a paragraph indent. To start a genuinely new paragraph instead, indented as the class prescribes, leave a blank line after `\z.:`

```
| \ex. An example. \z.  
|  
| A new, ordinarily indented paragraph.
```

renders as:

- (9) An example.

 A new, ordinarily indented paragraph.

3.4 Custom labels

An optional argument replaces the number, without advancing the counter — useful for repeating an earlier example, or for primed variants:

```
| \ex.[(7$' $)] A repeated or modified example.
```

renders as:

- (7') A repeated or modified example.

3.5 An environment interface

Everything above can equally be written with conventional environments, whose syntax is that of **gb4e**. They are enabled by a package option: `\usepackage[gb4e]{linguexx}` loads *only* this syntax — the dot commands are then not defined, which keeps documents and error messages clean, and as a side benefit leaves the kernel accent commands `\b`, `\c`, `\d` untouched, so no **hyperref** workaround is needed. The default (equivalently, the option **lazy**) loads only the dot syntax of the preceding sections. The two may be combined, `[lazy,gb4e]`, for a document that deliberately mixes them — this manual does — and then the two forms drive the same engine, sharing one counter, one label system and all layout parameters. These options choose the input *syntax*; a further, orthogonal option **legacy** chooses the *geometry* of the original **linguex** and may be added to either (section 8). **exe** is a *batch*: every `\ex` inside it — written *without* the period — is a new top-level example. **xlist** embeds a sub-level, nestable once more for the romans; its items are made by `\ex` (or by plain `\item`). A judgment is given in the **gb4e** manner as an optional argument, `\ex[⟨judgment⟩]{⟨text⟩}`, where the judgment may be *any* mark, not only the auto-detected four — or typed directly after a plain `\ex`, as everywhere else:

```

\begin{exe}
\ex\label{here} Here is one.
\ex[*]{Here another is.}
\ex Here are some with judgements.
\begin{xlist}
\ex[] {A grammatical sentence am I.}
\ex[??]{A dubious sentence is she.}
\end{xlist}
\end{exe}

```

renders as:

- (10) Here is one.
- (11) *Here another is.
- (12) Here are some with judgements.
 - a. A grammatical sentence am I.
 - b. ?? A dubious sentence is she.

Two differences follow from the syntax itself: the environments end at their `\end` and nothing else, so blank lines inside them are ordinary paragraph breaks; and `\z.` has no role to play, since an environment names its own end. An `xlist` may also sit inside a dot-command example, and `\a.` works inside `exe`. Use whichever form suits the document — or the editor.

3.6 Examples in footnotes

Inside a footnote, examples are numbered independently, in roman numerals, on their own counter; the running-text numbering is not disturbed, and references made inside the footnote resolve against the footnote’s series. Nothing special needs to be done: write `\ex.` as usual.¹

4 Grammaticality judgments

Marks are also given *spoken forms* for accessible PDFs, so that a screen reader announces a judgment by its meaning rather than by its glyph (section 4.4). That is one of the tagging features drawn together in section 10.

4.1 Automatic marks

The characters `*` and `?` and the commands `\#` and `\%`, in any combination, are recognized automatically when they open an example or sub-example. They are set in the margin, hanging to the left of the example text, so that judged and unjudged examples align:

¹Like so:

- (i) First footnote example.
- (ii) *Second one, with a judgment.

and (ii) refers to the second of these, not to a text example.

```

\ex.
\a. Ich habe geschlafen.
\b. ??Ich habe geschlaft.
\c. *Ich bin geschlafen.
\d. \#Ich schlafte.

```

renders as:

- (13) a. Ich habe geschlafen.
b. ??Ich habe geschlaft.
c. *Ich bin geschlafen.
d. #Ich schlafte.

The mark must come first in the item, *before* anything else including a `\label`. The conventional readings are the usual ones (* ungrammatical, ? degraded, # semantically or pragmatically deviant, % varying across speakers), but the package attaches no meaning to them: it only places them.

4.2 Space for judgments

A hanging mark takes no horizontal space, so it can never push the example text out of alignment — but it can, if long enough, run into the example *label* to its left. The room available is the clear space inside the label box plus the label gap, i.e.

$$\langle labelwidth \rangle - \text{width of the printed label} + \text{\Exlabelsep}.$$

At the main level the number box (`\Exlabelwidth`, 2.6em) leaves ample room. At the sub-levels the default widths (`\SubExlabelwidth` 1.6em, `\SubSubExlabelwidth` 1.9em) are chosen so that **two marks built from ? and * (??, ?*, ...)** fit clear of the letter. Note that `\#` and `\%` are nearly twice as wide as `?` in most fonts, so they count roughly double. If you use **three marks** (or wide marks) on sub-examples **as a routine matter**, widen the boxes:

```

\setlength{\SubExlabelwidth}{2.4em}
\setlength{\SubSubExlabelwidth}{2.5em}

```

which carries `?*#` comfortably. (An occasional long mark that grazes the letter is harmless; only alignment is guaranteed, and that unconditionally.)

4.3 Arbitrary marks: `\jdg`

For any other mark, `\jdg{<mark>}` hangs an arbitrary symbol into the margin, at any position, not only at the start of an example:

```

\ex. \jdg{\dag}A mark of one's own choosing.
\ex. \jdg{\to}Or an arrow, or anything else.

```

renders as:

- (14) † A mark of one's own choosing.
(15) → Or an arrow, or anything else.

Because the mark occupies no horizontal space, alignment is preserved whatever its width. To give a recurrent mark a name of its own, use `\DeclareJudgment{⟨command⟩}{⟨mark⟩}`:

```
| \DeclareJudgment{\dubious}{\dag\dag}
| \ex. \dubious A doubly daggered example.
```

renders as:

(16) †† A doubly daggered example.

The distance between mark and text is the length `\JdgSep` (default 0.15em).

4.4 Spoken forms for tagged PDFs

A judgment mark is a symbol whose meaning a screen reader cannot guess: read literally, `*` is “asterisk”. When the document is compiled with the PDF tagging code active (a `\DocumentMetadata` line enabling the tagging `testphase`), `linguexx` wraps each mark in a structure element carrying an *alternate text* so it is announced by its meaning instead. The built-in marks have sensible defaults:

mark	spoken as
*	ungrammatical
?	questionable
??	highly questionable
?*	extremely degraded
#	infelicitous
%	grammatical for some speakers

This affects only the tagged PDF’s accessibility layer; the printed output is unchanged, and nothing happens on an engine or in a document without active tagging. To set the spoken form of a mark, give `\DeclareJudgment` its optional `spoken` key — which both names the mark and registers its spoken form, so a leading scanned mark is announced the same way —

```
| \DeclareJudgment[spoken={marginal, some speakers}]{\pcent}{\%}
```

or, for a mark you do not need to name, `\SetJudgmentSpoken {⟨mark⟩}{⟨phrase⟩}`. An explicit `\jdg` may also carry a one-off spoken form as an optional argument: `\jdg[⟨ungrammatical⟩]{⟨*⟩}`.

5 Interlinear glosses

5.1 Two and three tiers

`\gll` takes an object line and a gloss line, each ended by `\\`; `\glll` takes three lines. `\glt` introduces the free translation, which is set as an ordinary line, not aligned:

```
| \ex. \gll Der Hund hat geschlafen \\
|         the dog has slept          \\
| \glt ‘The dog slept.’
```

renders as:

- (17) Der Hund hat geschlafen
the dog has slept
‘The dog slept.’

Words are separated by spaces, and *braced material counts as one word*. This is how one object-language word is glossed by several English words, or the reverse:

```
\ex. \gll Das {kleine Kind} schl\"aft \\
        the {little child} sleeps   \\
\glt ‘The little child sleeps.’
```

renders as:

- (18) Das kleine Kind schläft
the little child sleeps
‘The little child sleeps.’

If the tiers are of unequal length, the missing cells are simply left empty — no error. A gloss too long for the line wraps between columns, never inside one.

5.2 Any number of tiers: \gl ... \endgl

Heavily annotated data — transliteration, segmentation, gloss, category line — needs more than three tiers. \gl takes any number of lines, each ended by \\, up to \endgl:

```
\GlossTierFont{4}{\textit}

\ex. \gl Der Hund schl\"aft \\
        der Hund schlaf-t \\
        DEF.NOM.SG.M dog sleep-3SG \\
        D N V \\ \endgl
\glt ‘The dog sleeps.’
```

renders as:

- (19) Der Hund schläft
der Hund schlaf-t
DEF.NOM.SG.M dog sleep-3SG
D *N* *V*
‘The dog sleeps.’

All the conventions of \gll carry over: spaces separate words, braces group them, unequal tiers get empty cells, long examples wrap between columns. \gll and \glll are in fact abbreviations of this one engine.

5.3 Fonts and spacing

Each tier is set with a one-argument command, declared by \GlossTierFont{<n>}{<command>}. Tiers 1–3 have traditional names of their own — \eachwordone, \eachwordtwo, \eachwordthree — which may be redefined instead; all default to \textnormal, so glosses inherit the surrounding font (under beamer, they come out sans-serif, as they should). To set the gloss line in small capitals throughout:

```
| \renewcommand{\eachwordtwo}[1]{\textsc{#1}}
```

The space between columns is the macro `\GlossSep`, a glue specification (default `.5em` plus `.3em` minus `.1em`), changed with `\renewcommand`.

5.4 The language of a tier (tagged PDFs)

The object line of a gloss is usually in a different language from the document. In a tagged PDF that difference can be recorded so that a screen reader pronounces each tier with its own phonetics rather than reading, say, German words as though they were English. Declare the language of a tier with `\GlossTierLang{<tier>}{<code>}`, where `<code>` is a language tag such as `de` or `fr`:

```
| \GlossTierLang{1}{de}
| \ex.
| \gll Der Hund bellte.\\ the dog barked.\\
| \glt 'The dog barked.'
```

Each word of tier 1 is then wrapped, in the structure tree, in a span carrying that language; the other tiers are unaffected.

`\GlossTierLang` obeys the usual scope rule. Given in the preamble or in the document body between examples, it sets a *document-wide default* for that tier. Given *inside* an example, before the gloss, it overrides that tier for that one example only and reverts afterwards, so a document that glosses several object languages needs only a local `\GlossTierLang` in the odd example that departs from the default:

```
| \GlossTierLang{1}{de}           % default: object line is German
| ...
| \ex.
| \GlossTierLang{1}{fr}           % this example only
| \gll Le chien aboya.\\ the dog bark.\lpzg{pst}\\
| \glt 'The dog barked.'
```

As with the rest of the tagging support this is inert unless tagging is active (section 10): it changes nothing in the printed gloss and nothing in an untagged document. A tier with no declared language behaves exactly as before.

5.5 Gloss abbreviations: `\lpzg`

Category labels in a gloss are written in small capitals by convention (SG, PST, NOM). `\lpzg{<label>}` is a convenience for exactly this: it sets its argument in small capitals, so `\lpzg{<sg>}` is a shorter, meaning-bearing spelling of `\textsc{<sg>}`. A whole compound label goes in one call, written the Leipzig way — a leading person number flush against the number, then further categories separated by periods:

```
| \gll Der Hund bellte.\\
|         the dog \lpzg{3sg.pst}\\
| \glt 'The dog barked.'
```

renders as:

```
Der Hund bellte.
the dog   3SG.PST
'The dog barked.'
```

`\lpzg{<3sg.pst>}` simply sets 3SG.PST. Its argument is a label you would otherwise have typed by hand; there is no fixed vocabulary as far as *typesetting* is concerned. What `\lpzg` adds over `\textsc` appears only in a tagged PDF, where it records the spoken expansion of the label (“third person singular past”) for a screen reader; that behaviour, and the list of abbreviations it knows, are described in section 10.1. `\lpzg` is self-contained and does not require, or interact with, the `leipzig` package.

5.6 Abbreviations: `\exg.` and `\ag.`

`\exg.` abbreviates `\ex.` `\gll`, and `\ag.` through `\fg.` abbreviate `\a.` `\gll` and so on, for glossed sub-examples. Judgments still work in front of the gloss:

```
\exg. *Das Beispiel funktionieren \\
      the example work.INF         \\
\glt (intended: ‘The example works.’)
```

renders as:

(20) *Das Beispiel funktionieren
 the example work.INF
 (intended: ‘The example works.’)

6 Cross-references

6.1 Labels

Examples and sub-examples are ordinary L^AT_EX counters: `\label` them and `\ref` them. The reference prints with parentheses, and a sub-example reference includes its letter:

```
\ex.\label{ex:main}
\ a.\label{ex:a} Erste.
\ b.\label{ex:b} Zweite.

See \ref{ex:main}, and in particular \ref{ex:b}.
```

renders as:

(21) a. Erste.
 b. Zweite.

See (21), and in particular (21b).

6.2 References without parentheses

In running text one often wants “example 3 b” rather than “example (3 b)”. Every reference command has a p-prefixed twin that omits the parentheses: `\pref`, `\pNext`, `\pLast`, `\pNNext`, `\pLLast`, `\pTextNext`.

```
| \ref{ex:b} with parentheses, \pref{ex:b} without.
```

renders as:

(21b) with parentheses, 21b without.

6.3 Relative references

To refer to a neighbouring example without labelling it:

Command	Refers to
<code>\Next</code>	the next example
<code>\NNext</code>	the one after that
<code>\Last</code>	the previous example
<code>\LLast</code>	the one before that
<code>\TextNext</code>	the next example in the running text (for use inside footnotes)

Inside a footnote these refer to the footnote’s own series, which is nearly always what is meant; `\TextNext` escapes to the main series.

6.4 Ranges

`\refrange{⟨first⟩}{⟨last⟩}` sets a compact range over sub-examples, printing “(3a–c)” rather than “(3a)–(3c)”. The sub-examples must be labelled with `\sublabel` instead of `\label` (this records the letter in the `.aux` file; `\sublabel` also does everything `\label` does, so `\ref` keeps working):

```
\ex.  
\a.\sublabel{r:a} Erste.  
\b. Zweite.  
\c.\sublabel{r:c} Dritte.  
  
The examples in \refrange{r:a}{r:c} all show ...
```

renders as:

(22) a. Erste.
 b. Zweite.
 c. Dritte.

The examples in (22a–c) all show ...

`\prefrange` is the parenthesis-free variant; `\Refrange` sets a range over two whole examples, as (10)–(12). The range dash is the macro `\rangedash` (default an en-dash). The dash between number and letter is `\firstrefdash` — empty by default, so “(3a)”, but - under *legacy* — and between letter and roman numeral `\secondrefdash` (- in both modes); see section 8.

7 Further apparatus

7.1 Stacked alternatives: `\alt`

A set of interchangeable constituents is conventionally stacked inside a curly brace. `\alt` takes any number of brace groups and stacks them:

```
\ex. \alt{\sout{This girl in the red coat}}{She}{Mary} will put a  
     picture of \alt{Bill}{him} on your desk.
```

renders as:

$$(23) \quad \left\{ \begin{array}{c} \text{This girl in the red coat} \\ \text{She} \\ \text{Mary} \end{array} \right\} \text{ will put a picture of } \left\{ \begin{array}{c} \text{Bill} \\ \text{him} \end{array} \right\} \text{ on your desk.}$$

The alternatives are collected as long as brace groups follow one another immediately; the first non-brace token ends the list. Strike-through (`\sout`) is available for an excluded alternative if your document loads `ulem` — the package does not load it for you. An optional argument sets the alignment: `[c]` (default), `[l]`, `[r]`.

A name clash worth knowing. `beamer` uses `\alt` for overlays. The package detects this, leaves `beamer`’s command alone, and notes it in the log. **Under `beamer`, write `\lxAlt`**; everywhere else `\alt` and `\lxAlt` are the same command.

For a light word-level alternation inside a glossed line a slash (`gehe/laufe`) is often enough; when each alternative needs a gloss of its own, use `\altg` (section 7.2).

`\alt` is set in text mode: the alternatives are the rows of a `tabular`, braced on both sides with braces drawn with `TikZ`, so no mathematics is involved (the package needs `graphicx` and `tikz`, not `amsmath`). The braces are tunable through four macros — `\AltBraceWidth` (default `0.9ex`), `\AltBraceAmplitude` (`4pt`, how far the brace’s centre point juts out), `\AltBraceSep` (`0.3em`, the gap between brace and stack), and `\AltBraceRaise` (`0pt`, a vertical nudge) — redefined with `\renewcommand`.

Because the alternatives are ordinary text rather than a formula, they are read sensibly in a tagged PDF. Under active tagging `\alt` wraps the stack in a span carrying a spoken form of the list — “This girl in the red coat, She, or Mary” — built from the alternatives themselves, with any formatting (such as the `\sout` above) stripped for speech. The printed output is unchanged, and an untagged document is unaffected (section 10).

7.2 Glossed alternatives: `\altg`

Where `\alt` stacks bare constituents, `\altg` stacks a paradigm in which each alternative carries its own gloss. Inside an interlinear gloss it is written *twice* — once in the object line with the object words, once in the gloss line with their glosses:

```
\exg. Die \altg{Frau}{Socke}{Maus}{Tonne} ist hier.\
      The.\lpzg{sg} \altg{woman.\lpzg{sg}}{sock.\lpzg{sg}}%
      {mouse.\lpzg{sg}}{ton.\lpzg{sg}} is.\lpzg{prs} here.\
```

renders as:

$$(24) \quad \begin{array}{c} \text{Die} \\ \text{The.SG} \end{array} \left\{ \begin{array}{cc} \text{Frau} & \text{woman.SG} \\ \text{Socke} & \text{sock.SG} \\ \text{Maus} & \text{mouse.SG} \\ \text{Tonne} & \text{ton.SG} \end{array} \right\} \begin{array}{cc} \text{ist} & \text{hier.} \\ \text{is.PRS} & \text{here.} \end{array}$$

The two calls occupy the two tiers of one gloss column and assemble a single paradigm: object stack on the left, gloss stack to its right (offset by `\AltgColSep`), braced on *both* sides. The block is centred on the midline between the object tier and the gloss tier: with four alternatives, rows two and three ride the object and gloss lines and the outer rows protrude symmetrically, with the surrounding lines keeping clear. The example number stays on the object baseline, where `\exg.` puts it.

Three rules of use. The two calls must list the same number of alternatives — the package stops with an error otherwise. No spaces may separate the brace groups: a space both ends the collection and splits the gloss into separate columns; break long calls across source lines with %, as above. And the alternatives are paired by position — the n -th gloss glosses the n -th object word.

Outside a gloss, a single `\altg` sets one both-braced stack on the current baseline — `\alt` with a closing brace, in effect.

Because each stack already competes for both tiers, `\lpzg` inside it prints as plain small caps, without the abbreviation span of section 10; the expansions are not lost — they reappear in the spoken forms. Under active tagging each call is wrapped in a span carrying a spoken `/Alt` of its own list — “Frau, Socke, Maus, or Tonne” for the object call, “woman.singular, sock.singular, mouse.singular, or ton.singular” for the gloss call — with simple `\lpzg` keys expanded from the Leipzig table; compound keys (`3sg.pst`) and unknown keys are spoken as printed. Like `\alt`, none of this is mathematics, so the alternatives remain ordinary tagged text.

Two settings of its own, redefined with `\renewcommand`: `\AltgColSep` (default 1.2em), the gap between the object and gloss stacks, and `\AltgTransFont` (default `\normalfont`), the font of the gloss stack. The braces are drawn with the `\alt` tunables (`\AltBraceWidth`, `\AltBraceAmplitude`, `\AltBraceSep`). If some other package owns `\altg`, the package leaves it alone and only `\lxAltg` is available, as with `\alt`/`\lxAlt`.

7.3 Source attributions: `\exsource`

`\exsource` sets an attribution flush right at the end of an example, dropping it onto a line of its own if it does not fit:

```
\ex. This girl in the red coat will put a picture of Bill on your desk.
\exsource{(Perlmutter 1971: 76)}
```

renders as:

(25) This girl in the red coat will put a picture of Bill on your desk. (Perlmutter 1971: 76)

It works in every part of a multipart example, and after a gloss. The font is the hook `\ExSourceFont` (default `\normalfont\footnotesize`); the argument may contain a `\cite`.

8 Layout

The geometry is uniform across levels: each level reserves a label box of fixed width, followed by the shared gap `\Exlabelsep`; the text margin of a level is its label width plus `\Exlabelsep`, measured from the margin of the level above (figure 1). Judgments hang to the left of the text margin and take no space.

Every parameter is an ordinary length, set with `\setlength` anywhere in the document:

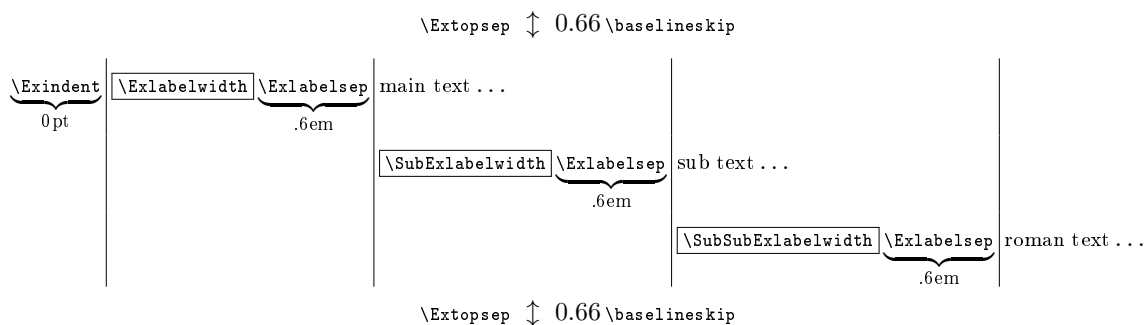


Figure 1: The default box model. On each line, reading left to right: an optional `\Exindent` (0 pt), the label box, then the shared gap `\Exlabelsep`; the text of a level begins one label box plus one `\Exlabelsep` to the right of the level above, so the boxes cascade down the page. Each box is named for the length that fixes its width (`\Exlabelwidth` 2.6 em, `\SubExlabelwidth` 1.6 em, `\SubSubExlabelwidth` 1.6 em) and holds the number, the letter and the roman label respectively; the vertical rules mark the three text margins. `\Extopsep` is the space left above and below. Under `legacy` the sub-levels are driven instead by two leftmargins, `\SubExleftmargin` and `\SubSubExleftmargin`, as in `linguex`.

Length	Default	Role
<code>\Extopsep</code>	<code>.66\baselineskip</code>	Vertical space before and after an example.
<code>\Exredux</code>	<code>-.66\baselineskip</code>	Correction inserted between <i>consecutive</i> examples, so the gap is not doubled. Keep it at <code>-\Extopsep</code> for an even rhythm.
<code>\Exlabelwidth</code>	2.6em	Width of the number box — room for “(199)”. Widen it for four-digit numbering.
<code>\Exlabelsep</code>	.6em	Gap between any label box and its text.
<code>\SubExlabelwidth</code>	1.6em	Width of the letter box (“a.”). Deliberately wider than the letter: the surplus is where a hanging judgment goes (section 4.2).
<code>\SubSubExlabelwidth</code>	1.6em	Width of the roman box, sized for the widest label up to “vi.” (“iii.”), which is the same width as the widest letter; equal to <code>\SubExlabelwidth</code> , so the two sub-levels share one box width.
<code>\JdgSep</code>	0.15em	Gap between a hanging judgment and the text.

A few further knobs are macros, not lengths, and are changed with `\renewcommand`: `\GlossSep` (glue between gloss columns) and the three reference dashes. `\rangedash` (default `--`) separates the two ends of a range; `\firstrefdash` sits between an example number and a sub-example letter in a reference such as “(3a)”, and `\secondrefdash` between the letter and a roman numeral in “(3a-i)”. Their defaults differ by mode: in the

default scheme `\firstrefdash` is *empty* (giving “(3a)”) and `\secondrefdash` is `-`; under **legacy** both are `-`, so the same reference prints “(3-a)” and “(3-a-i)”, as `linguex` does. The sub-label delimiters `\SubExLBr/\SubExRBr` and `\SubSubExLBr/\SubSubExRBr` bracket the letter and roman labels themselves; they are `{}/{}.}` in both modes for the letter (“a.”), while the roman is bare “i.” by default but parenthesised “(i)” under **legacy**.

The counters are `ExNo`, `SubExNo`, `SubSubExNo` and `FnExNo` (footnotes). To change the style of the numbering, redefine `\theExNo` and its relatives in the usual L^AT_EX way.

The default scheme and legacy. The values in the table are the *default* scheme. The option **legacy** selects instead the exact geometry and conventions of `linguex`: the number box is sized per example from the width of its digits rather than fixed at `\Exlabelwidth`; the sub-levels are positioned by two leftmargins, `\SubExleftmargin (2em)` and `\SubSubExleftmargin (2.4em)`, rather than by a label width plus `\Exlabelsep`; roman sub-sub-labels print as “(i)”; `\firstrefdash` is `-`; and, in a book-class document, `ExNo` resets at each `\chapter`. **legacy** is orthogonal to the choice of input syntax: combine it freely, as in `[legacy,gb4e]`. Both parameter sets exist in either mode — `\SubExleftmargin` always equals `\SubExlabelwidth + \Exlabelsep` — so a document can be nudged from one scheme toward the other one length at a time.

`\resetExdefaults` restores every layout parameter to the values of the mode in force; it is what the package runs for you as it loads. Because the defaults are applied at load time, not `\AtBeginDocument` as in `linguex`, a `\setlength` in your preamble simply wins — there is no need to hook it into `\resetExdefaults` to keep it from being overwritten. The two font-relative lengths, `\Extopsep` and `\Exredux`, are the exception: they are finalised at `\begin{document}` so that they follow the body font, unless you have already given either an explicit value.

9 Command reference

Command	Effect
<i>Examples</i>	
<code>\ex.</code>	Start a numbered example; a blank line ends it. <code>\ex.[{label}]</code> uses a custom label without advancing the counter.
<code>\a.</code>	Open a level of sub-examples (letters, then roman).
<code>\b. - \f.</code>	Next item at the level currently open.
<code>\z.</code>	Leave one level; from the letter level or an example without sub-examples, end it.
<code>exe, xlist</code>	gb4e -compatible batch and sub-level environments (package option gb4e); items via <code>\ex[{judgment}]{text}</code> , or plain <code>\item</code> .
<code>\exg.</code>	<code>\ex.</code> followed by <code>\gll</code> .
<code>\ag. - \fg.</code>	<code>\a.-\f.</code> followed by <code>\gll</code> .
<i>Judgments</i>	
<code>\jdg{mark}</code>	Hang an arbitrary mark in the margin.

continued on the next page

Command	Effect
<code>\DeclareJudgment</code> <code>{⟨cmd⟩}{⟨mark⟩}</code>	Give a mark a name.
<i>Glosses</i>	
<code>\gll</code> , <code>\glll</code>	Two-, three-tier gloss; lines end in <code>\\</code> .
<code>\gl ... \endgl</code>	Gloss with any number of tiers.
<code>\glt</code>	Free-translation line. (<code>\gln</code> is a synonym.)
<code>\GlossTierFont</code> <code>{⟨n⟩}{⟨cmd⟩}</code>	Font command for tier <code>⟨n⟩</code> .
<code>\GlossTierLang</code> <code>{⟨n⟩}{⟨code⟩}</code>	Language of tier <code>⟨n⟩</code> for tagged PDFs (section 5.4).
<code>\lpzg{⟨abbr⟩}</code>	Gloss abbreviation in small caps, with spoken expansion when tagged (section 5.5).
<code>\SetLeipzig</code> <code>{⟨abbr⟩}{⟨expansion⟩}</code>	Add or override a Leipzig abbreviation expansion.
<code>\eachwordone/two/three</code>	Font commands for tiers 1–3.
<code>\GlossSep</code>	Glue between gloss columns.
<i>References</i>	
<code>\label</code> , <code>\ref</code>	As usual; references print in parentheses.
<code>\sublabel{⟨key⟩}</code>	Label a sub-example for use in a range.
<code>\Next</code> , <code>\NNext</code> , <code>\Last</code> , <code>\LLast</code>	Relative references.
<code>\TextNext</code>	Next example of the running text (from a footnote).
<code>\pref</code> , <code>\pNext</code> , ...	Parenthesis-free twins of all of the above.
<code>\refrange{⟨a⟩}{⟨b⟩}</code>	Compact range, as (3a–c).
<code>\prefrange</code> , <code>\Refrange</code>	Without parentheses; over whole examples.
<i>Further apparatus</i>	
<code>\alt{⟨...⟩}{⟨...⟩}</code>	Stacked alternatives; [<code>⟨c/l/r⟩</code>] alignment. Under beamer: <code>\lxAlt</code> .
<code>\altg{⟨alt⟩}{⟨alt⟩}...</code>	Glossed alternatives: written in both lines of <code>\exg</code> . (objects, then glosses); both-braced, centred. Also <code>\lxAltg</code> .
<code>\exsource{⟨text⟩}</code>	Flush-right attribution.
<i>Layout and options</i>	
<code>lazy</code> , <code>gb4e</code>	Package options: dot syntax (default), <code>gb4e</code> environment syntax; combinable.
<code>legacy</code>	Package option: the geometry and dash conventions of <code>linguex</code> ; orthogonal to the syntax options.
<code>\resetExdefaults</code>	Restore all layout lengths to the current mode’s defaults.

continued on the next page

Command	Effect
<code>\firstrefdash</code>	Number–letter dash in “(3a)”; empty by default, - under legacy .
<code>\secondrefdash</code>	Letter–roman dash in “(3a-i)”; - in both modes.
<code>\rangedash</code>	Dash in a whole-example range.

10 PDF tagging and accessibility

linguexx is aware of the L^AT_EX project’s PDF tagging code. If the document begins with a `\DocumentMetadata` line that turns tagging on, for example, on a current T_EX distribution,

```
| \DocumentMetadata{lang=en,tagging=on}
```

then examples are written into the PDF’s structure tree as genuine, accessible objects rather than as loose ink. Without such a line nothing below applies: the printed output is identical and the package behaves exactly as in the rest of this manual.

The exact spelling of that line is a moving target. The `tagging=on` key is the recommended form from the L^AT_EX release of November 2025 onward. On distributions between roughly 2023 and 2025 the tagging code lives behind the experimental `testphase` key instead; the portable choice there, which also works on current T_EX, is

```
| \DocumentMetadata{lang=en,testphase={phase-III}}
```

Both enable the same comprehensive tagging as far as linguexx is concerned. Avoid the older habit of naming individual modules (`testphase={tagpdf,...}`): those names have been reorganised and can now load only part of the machinery.

What works today, on pdfL^AT_EX, X_YL^AT_EX and LuaL^AT_EX alike. The structure tree is valid, including the awkward case of an example inside a footnote. Each example is a list item with its number as the label and its content as the body; the letter and roman sub-levels are nested lists inside it, so assistive technology can walk into and out of the hierarchy. Those lists are marked as *ordered*, matching their numbering. Grammaticality marks are given a spoken form, so a screen reader announces “ungrammatical” rather than “asterisk” (section 4.4). In an interlinear gloss each column — an object word together with its aligned gloss(es) — is grouped as one structure element, so the gloss is read and navigated word bundle by word bundle, in the object-then-gloss order, rather than as one undifferentiated run of text. The object language of a tier can be recorded (section 5.4), so it is pronounced with its own phonetics; and category abbreviations written with `\lpzg` carry their expansion (section 10.1), so a reader hears “past” rather than “pee-ess-tee”. Stacked alternatives (`\alt`) are set in text mode rather than as a formula, and carry a spoken form of the list (section 7), so they are read as “A, B, or C” rather than as a run of braces and glyphs.

Validation. A document built with the accessibility preamble above validates as PDF/UA-2: veraPDF reports it conformant, with every rule and check passing and no exceptions. Conformance is a document-level property, not only a matter of the examples — it also needs a document title and a few other pieces of metadata — so the package ships a short checklist (PDFUA-CHECKLIST) giving the preamble that supplies them. What a validator cannot certify is that the result is *pleasant* to listen to; that still wants testing with a real screen reader.

A caution on stability. Tagging in L^AT_EX is, at the time of writing, an evolving feature: the interface has already moved from named `testphase` modules through `testphase={phase-III}` to the `tagging=on` key, and the internal machinery still changes between releases. A recent L^AT_EX is required. `linguexx`’s support is written to degrade quietly — if the tagging machinery is absent or turned off, the relevant code does nothing — but it should be regarded as provisional and may need a touch-up as the tagging project matures. The `legacy` option is orthogonal to all of this; tagging support targets the default mode.

10.1 Gloss abbreviations and their expansions

Under active tagging, `\lpzg{<label>}` does more than set small capitals (section 5.5): it records the label’s *expansion* as the PDF “expansion text” (the `/E` entry of an abbreviation), so a screen reader announces “singular” while the page still shows `sg` and copy-and-paste still yields `sg`. This is the correct mechanism for an abbreviation — distinct from an *alternate text*, and unlike `/ActualText` it leaves the copied text alone.

A compound label is parsed for the expansion: it is split on periods, a leading 1, 2 or 3 is taken as the person, and each piece is expanded and joined into one phrase, so `\lpzg{3sg.pst}` reads “third person singular past”. A piece not in the table below passes through verbatim; a label with nothing recognisable in it gets no expansion, and no warning, so project-specific labels are safe. Extend or override the table with `\SetLeipzig{<label>}{<expansion>}`:

```
| \SetLeipzig{obv}{obviative}
```

The built-in table is the standard Leipzig Glossing Rules list; the meaning column is what a screen reader speaks.

label	meaning	label	meaning
1	first person	INDF	indefinite
2	second person	INF	infinitive
3	third person	INS	instrumental
A	agent	INTR	intransitive
ABL	ablative	IPFV	imperfective
ABS	absolutive	IRR	irrealis
ACC	accusative	LOC	locative
ADJ	adjective	M	masculine
ADV	adverbial	N	neuter
AGR	agreement	NEG	negative
ALL	allative	NMLZ	nominalizer
ANTIP	antipassive	NOM	nominative
APPL	applicative	OBJ	object
ART	article	OBL	oblique
AUX	auxiliary	P	patient
BEN	benefactive	PASS	passive
CAUS	causative	PFV	perfective
CLF	classifier	PL	plural
COM	comitative	POSS	possessive
COMP	complementizer	PRED	predicative
COMPL	completive	PRF	perfect
COND	conditional	PROG	progressive
COP	copula	PROH	prohibitive

label	meaning	label	meaning
CVB	converb	PROX	proximal
DAT	dative	PRS	present
DECL	declarative	PST	past
DEF	definite	PTCP	participle
DEM	demonstrative	PURP	purposive
DET	determiner	Q	question particle
DIST	distal	QUOT	quotative
DISTR	distributive	RECP	reciprocal
DU	dual	REFL	reflexive
DUR	durative	REL	relative
ERG	ergative	RES	resultative
EXCL	exclusive	S	argument of intransitive verb
F	feminine	SBJ	subject
FOC	focus	SBJV	subjunctive
FUT	future	SG	singular
GEN	genitive	TOP	topic
IMP	imperative	TR	transitive
INCL	inclusive	VOC	vocative
IND	indicative		

11 Notes and limitations

- **Two sub-levels.** A third `\a.` is an error. Deeper hierarchies are almost always better expressed as separate examples.
- **The period is part of the command.** `\ex.`, `\a.`, `\z.` — not `\ex`, `\a`, `\z`. This is what allows the syntax to be so light.
- **`\z.` at brace depth 0.** `\z.` is recognized in the body of an example, not inside a brace group within it.
- **ulem is not loaded.** For struck-through alternatives (`\sout` inside `\alt`), load `ulem` yourself; the usual choice is `\usepackage[normalem]{ulem}`, which keeps `\emph` italic.
- **hyperref and `\b`, `\c`, `\d`.** Under the default `lazy` mode, `linguexx` defines `\b.`, `\c.`, `\d.` and re-asserts them at `\begin{document}`, since `hyperref` redefines these names (they are accent commands used in bookmarks); either load order works, but the bar-under, cedilla and dot-under accents are then unavailable. Under `[gb4e]` alone the letter commands are never defined and the accents work normally.
- **Footnote examples.** End an example inside a footnote with a blank line or `\z.` rather than letting it run into the footnote’s own end.
- **Glosses and alternatives do not combine.** See section 7.
- **Where `expex` still does more.** Tier *count* and tier *fonts* are unrestricted here, but the vertical layout of a gloss is not individually adjustable per tier, and there is no `keyval` interface. See section 2.

linguexx owes its input syntax to Wolfgang Sternefeld's *linguex*, its glossing commands to the *cgloss4e* of Hans-Peter Kolb and Craig Thiersch, the ambition of its gloss engine to John Frampton's *expex*, and its parenthesis-free references to a construction of Alan Munn's. The implementation is its own; the debts are not.